High-Order Explicit Manufactured FDA Solver Verification Report

Generated report for `highOrderManufacturedFDAExplicit`

May 7, 2026

Abstract

This report documents the current `highOrderManufacturedFDAExplicit` solver. The solver advances a monodomain-like manufactured activation equation coupled to local state variables. The transmembrane potential can be diffused with either the standard finite-volume operator or an LRE high-order flux operator. The nonlinear ionic current can be evaluated at cell centres or volume-integrated from dedicated cell quadrature points, where the local ODE states are also evolved. The convergence tables and plotting script read the current two- and three-dimensional data in `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAExplicit/results/example0/2D` and `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAExplicit/results/example0/3D`.

1 Manufactured FDA problem

The unknowns are the transmembrane potential V_m and state variables u_1, u_2, and u_3. The manufactured problem is

$$\frac{\partial V_m}{\partial t} = \frac{1}{\chi C_m}\nabla \cdot (D\nabla V_m) - I_{\text{ion}}(V_m, u_1, u_2, u_3), \tag{1}$$

$$\frac{\partial u_1}{\partial t} = (u_1 + u_3 - V_m)^2 u_2^2 + \frac{1}{2}(u_1 + u_3 - V_m)u_2^2(V_m - u_3), \tag{2}$$

$$\frac{\partial u_2}{\partial t} = -(u_1 + u_3 - V_m)u_2^3, \tag{3}$$

$$\frac{\partial u_3}{\partial t} = 0. \tag{4}$$

The ionic current is

$$I_{\text{ion}} = -\frac{1}{2}(u_1 + u_3 - V_m)u_2^2(V_m - u_3) + \frac{\beta}{\chi C_m}(V_m - u_3), \tag{5}$$

where β is the exact Laplacian eigenvalue associated with the selected manufactured mode and conductivity tensor.

For the two-dimensional runs used in this report,

$$F(x, y) = \cos(\pi x)\cos(2\pi y), \tag{6}$$

$$G(x, y) = 1 + xy^2, \tag{7}$$

$$V_m^\star(x, y, t) = \sqrt{1 + t}\, F(x, y), \tag{8}$$

$$u_1^\star(x, y, t) = (1 + t)G(x, y) + V_m^\star(x, y, t), \tag{9}$$

$$u_2^\star(x, y, t) = \frac{1}{(1 + t)\sqrt{G(x, y)}}, \tag{10}$$

$$u_3^\star = 0. \tag{11}$$

The code also contains one- and three-dimensional manufactured definitions.

2 Spatial discretization

2.1 Diffusion of V_m

The potential equation can use the standard OpenFOAM finite-volume Laplacian or a high-order flux computed by LRE. In the high-order case, local Taylor reconstructions are built from cell stencils. The options `p1`, `p2`, and `p3` use derivatives through first, second, and third order respectively. The diffusive flux through a face f is approximated by

$$\Phi_f \approx |S_f| \sum_{q=1}^{N_f} \omega_{fq} \, \mathbf{n}_f \cdot D_{o(f)} \nabla_h V_m(\mathbf{x}_{fq}), \tag{12}$$

where $o(f)$ is the owner cell and $\mathbf{x}_{fq}$ are face quadrature points. The cell Laplacian is then recovered from the divergence of these numerical fluxes.

2.2 Volume integration of I_{ion}

The current solver uses `useHighOrder_Iion` for the nonlinear source. The older input name `useHighOrder_states`, and the result-folder token `hoStates`, are still accepted for backward compatibility, but they should now be read as the quadrature level used for I_{ion} and for the local state values stored at those quadrature points.

If `useHighOrder_Iion` is off, the source and state ODEs are evaluated at cell centres:

$$I^n_{\text{ion},K} = I_{\text{ion}}\big(V^n_{m,K}, u^n_{1,K}, u^n_{2,K}, u^n_{3,K}\big) \,. \tag{13}$$

If `useHighOrder_Iion` is on, the ODE states are stored and advanced at the cell quadrature points $\mathbf{x}_{Kq}$, and the cell source is the quadrature average

$$I^n_{\text{ion},K} \approx \frac{1}{|K|} \sum_{q=1}^{N_K} \omega_{Kq} \, I_{\text{ion}}(V^n_m(\mathbf{x}_{Kq}), u^n_1(\mathbf{x}_{Kq}), u^n_2(\mathbf{x}_{Kq}), u^n_3(\mathbf{x}_{Kq})) \,. \tag{14}$$

The value $V_m(\mathbf{x}_{Kq})$ is obtained with the LRE reconstruction when `useHighOrder_Vm` is on; otherwise it is evaluated from a linear cell-centred reconstruction. The state fields written at cell centres are volume averages of the quadrature-point states,

$$u^n_{i,K} \approx \frac{1}{|K|} \sum_{q=1}^{N_K} \omega_{Kq} u^n_i(\mathbf{x}_{Kq}), \qquad i = 1, 2, 3. \tag{15}$$

3 Boundary conditions

For the standard finite-volume operator, the OpenFOAM patch fields are used directly. For the high-order diffusive flux, the solver now treats `zeroGradient` and `empty` patches as no-flux patches by setting the high-order boundary flux to zero:

$$\int_{\Gamma_N \cap \partial K} \mathbf{n} \cdot D \nabla V_m \, dS = 0. \tag{16}$$

This is the discrete high-order counterpart of the physical/electrophysiological no-flux condition. Fixed-value patches keep their boundary contribution in the LRE flux reconstruction. Thus the zero normal flux is imposed at the high-order flux level rather than only by the cell-centred OpenFOAM boundary-field type.

4 Explicit time integration

At each time step, all right-hand-side terms are evaluated from the known state at t^n, and the solver applies forward Euler:

$$V_{m,K}^{n+1} = V_{m,K}^{n} + \Delta t \left[\frac{1}{\chi C_m} L_h(V_m^n)_K - I_{\mathrm{ion},K}^{n} \right], \tag{17}$$

$$u_i^{n+1}(\mathbf{x}_{Kq}) = u_i^n(\mathbf{x}_{Kq}) + \Delta t \, R_i(V_m^n(\mathbf{x}_{Kq}), u^n(\mathbf{x}_{Kq})), \qquad i = 1, 2, 3, \tag{18}$$

when high-order ionic-current quadrature is active. In the cell-centred source mode, the same explicit update is applied to the cell-centred states.

5 Error measures and convergence rates

The solver writes cell-centred volume-weighted L_1, L_2, and L_∞ errors for V_m, u_1, and u_2. It also writes reconstructed quadrature errors, denoted by the superscript G. Pairwise rates use consecutive meshes,

$$p_{N_1,N_2} = \frac{\log(E_{N_1}/E_{N_2})}{\log(h_{N_1}/h_{N_2})}, \qquad h_N = 1/N. \tag{19}$$

The summary table reports the all-mesh log–log fit contained in each `spatial_convergence_rate_dt_*_total.dat` file.

6 Two-Dimensional Results

Table 1 reports the all-mesh fitted convergence orders obtained from the current two-dimensional manufactured runs in `results/example0/2D`. These are the 2D results only; the three-dimensional data are reported separately in the next section.

Table 1: Explicit solver: observed all-mesh convergence orders from the current 2D manufactured results. The Exp. column gives the expected pair V_m/I_{ion}: 2 for `NO` and `p1`, 3 for `p2`, and 4 for `p3`. The token `hoStates` in directory names denotes the high-order quadrature level used for I_{ion}.

V_m	I_{ion}	Δt	Exp.	$V_m^{L_2}$	$V_m^{G_2}$	$u_1^{L_2}$	$u_1^{G_2}$	$u_2^{L_2}$	$u_2^{G_2}$
`NO`	`NO`	10^{-3}	2/2	1.863	1.863	0.467	0.467	0.008	0.008
`NO`	`NO`	10^{-4}	2/2	1.982	1.982	1.990	1.990	0.303	0.303
`NO`	`NO`	10^{-5}	2/2	1.996	1.996	2.029	2.029	1.448	1.448
`NO`	`NO`	10^{-6}	2/2	1.997	1.997	2.000	2.000	1.981	1.981
`NO`	`p1`	10^{-3}	2/2	1.864	1.864	2.018	2.087	0.333	0.434
`NO`	`p1`	10^{-4}	2/2	1.982	1.982	2.001	2.071	1.730	1.668
`NO`	`p1`	10^{-5}	2/2	1.995	1.995	2.000	2.052	2.026	2.053
`NO`	`p1`	10^{-6}	2/2	1.997	1.997	1.999	2.050	2.001	2.054
`p1`	`NO`	10^{-3}	2/2	1.911	2.143	1.544	1.544	0.284	0.284
`p1`	`NO`	10^{-4}	2/2	1.912	2.146	1.616	1.616	1.376	1.376
`p1`	`NO`	10^{-5}	2/2	1.912	2.147	1.616	1.616	1.622	1.622
`p1`	`NO`	10^{-6}	2/2	1.912	2.147	1.616	1.616	1.626	1.626
`p1`	`p1`	10^{-3}	2/2	1.911	2.143	2.015	2.164	0.490	0.864
`p1`	`p1`	10^{-4}	2/2	1.912	2.146	1.999	2.157	1.716	2.006
`p1`	`p1`	10^{-5}	2/2	1.912	2.147	1.997	2.152	1.845	2.158
`p1`	`p1`	10^{-6}	2/2	1.912	2.147	1.997	2.152	1.837	2.158
`p2`	`NO`	10^{-3}	3/2	2.094	2.571	1.496	1.496	0.114	0.114
`p2`	`NO`	10^{-4}	3/2	2.133	2.604	2.235	2.235	0.988	0.988
`p2`	`NO`	10^{-5}	3/2	2.137	2.607	2.163	2.163	2.038	2.038
`p2`	`NO`	10^{-6}	3/2	2.137	2.608	2.154	2.154	2.152	2.152
`p2`	`p1`	10^{-3}	3/2	2.116	2.614	2.023	2.149	0.175	0.432
`p2`	`p1`	10^{-4}	3/2	2.159	2.649	2.005	2.787	1.517	1.603
`p2`	`p1`	10^{-5}	3/2	2.163	2.652	2.003	2.738	2.164	2.631
`p2`	`p1`	10^{-6}	3/2	2.164	2.653	2.003	2.731	2.106	2.732

V_m	I_{ion}	Δt	Exp.	V_m^{L2}	V_m^{G2}	u_1^{L2}	u_1^{G2}	u_2^{L2}	u_2^{G2}
p2	p2	10^{-3}	3/3	2.180	2.717	2.014	2.403	0.491	0.611
p2	p2	10^{-4}	3/3	2.237	2.758	2.004	2.980	1.526	1.887
p2	p2	10^{-5}	3/3	2.243	2.762	2.003	2.957	2.045	2.866
p2	p2	10^{-6}	3/3	2.244	2.762	2.003	2.953	2.103	2.950
p3	NO	10^{-3}	4/2	2.009	2.488	0.518	0.518	0.010	0.010
p3	NO	10^{-4}	4/2	2.167	2.646	2.067	2.067	0.325	0.325
p3	NO	10^{-5}	4/2	2.185	2.665	2.243	2.243	1.484	1.484
p3	NO	10^{-6}	4/2	2.187	2.667	2.204	2.204	2.173	2.173
p3	p1	10^{-3}	4/2	2.085	2.670	2.021	0.958	0.005	0.065
p3	p1	10^{-4}	4/2	2.289	2.873	2.003	2.431	1.324	0.638
p3	p1	10^{-5}	4/2	2.312	2.896	2.001	2.906	2.072	1.872
p3	p1	10^{-6}	4/2	2.314	2.898	2.001	2.870	1.997	2.774
p3	p3	10^{-3}	4/4	3.143	3.810	2.010	1.304	0.411	0.153
p3	p3	10^{-4}	4/4	2.414	3.137	2.000	2.628	1.390	0.909
p3	p3	10^{-5}	4/4	2.373	3.095	2.000	3.331	1.926	2.245
p3	p3	10^{-6}	4/4	2.369	3.091	1.999	3.425	1.990	3.319

6.1 2D Figures

7 Three-Dimensional Results

Table 2 reports the analogous all-mesh fitted convergence orders obtained from the current three-dimensional manufactured runs in `results/example0/3D`.

Table 2: Explicit solver: observed all-mesh convergence orders from the current 3D manufactured results. The Exp. column gives the expected pair V_m/I_{ion}: 2 for `NO` and `p1`, 3 for `p2`, and 4 for `p3`. The token `hoStates` in directory names denotes the high-order quadrature level used for I_{ion}.

V_m	I_{ion}	Δt	Exp.	V_m^{L2}	V_m^{G2}	u_1^{L2}	u_1^{G2}	u_2^{L2}	u_2^{G2}
NO	NO	10^{-3}	2/2	1.972	1.972	2.088	2.088	0.149	0.149
NO	NO	10^{-4}	2/2	1.978	1.978	1.993	1.993	1.698	1.698
NO	NO	10^{-5}	2/2	1.979	1.979	1.980	1.980	1.975	1.975
NO	NO	10^{-6}	2/2	1.979	1.979	1.978	1.978	1.979	1.979
NO	p1	10^{-3}	2/2	1.971	1.971	1.988	2.102	1.926	2.080
NO	p1	10^{-4}	2/2	1.977	1.977	1.988	2.099	1.986	2.105
NO	p1	10^{-5}	2/2	1.978	1.978	1.988	2.098	1.988	2.105
NO	p1	10^{-6}	2/2	1.978	1.978	1.988	2.098	1.988	2.105
p1	NO	10^{-3}	2/2	2.005	2.063	2.028	2.028	0.862	0.862
p1	NO	10^{-4}	2/2	2.007	2.063	1.994	1.994	1.959	1.959
p1	NO	10^{-5}	2/2	2.007	2.063	1.991	1.991	1.989	1.989
p1	NO	10^{-6}	2/2	2.007	2.063	1.991	1.991	1.990	1.990
p1	p1	10^{-3}	2/2	2.005	2.063	1.989	2.115	1.942	2.112
p1	p1	10^{-4}	2/2	2.007	2.063	1.989	2.114	1.994	2.125
p1	p1	10^{-5}	2/2	2.008	2.063	1.989	2.114	1.996	2.125
p1	p1	10^{-6}	2/2	2.008	2.063	1.989	2.114	1.996	2.125
p2	NO	10^{-3}	3/2	1.872	3.216	1.894	1.894	1.138	1.138
p2	NO	10^{-4}	3/2	1.873	3.216	1.870	1.870	1.859	1.859
p2	NO	10^{-5}	3/2	1.874	3.216	1.867	1.867	1.872	1.872
p2	NO	10^{-6}	3/2	1.874	3.216	1.867	1.867	1.872	1.872
p2	p1	10^{-3}	3/2	1.961	3.268	2.018	3.220	2.421	3.108
p2	p1	10^{-4}	3/2	1.963	3.269	2.017	3.218	3.134	3.203
p2	p1	10^{-5}	3/2	1.963	3.269	2.017	3.218	3.181	3.204
p2	p1	10^{-6}	3/2	1.963	3.269	2.017	3.218	3.185	3.204
p2	p2	10^{-3}	3/3	2.019	3.288	2.007	3.318	2.578	3.223
p2	p2	10^{-4}	3/3	2.021	3.289	2.006	3.316	3.203	3.299
p2	p2	10^{-5}	3/3	2.021	3.289	2.006	3.316	3.258	3.300
p2	p2	10^{-6}	3/3	2.021	3.289	2.006	3.316	3.263	3.300
p3	NO	10^{-3}	4/2	2.474	3.627	2.551	2.551	0.556	0.556
p3	NO	10^{-4}	4/2	2.478	3.629	2.476	2.476	2.347	2.347
p3	NO	10^{-5}	4/2	2.479	3.629	2.467	2.467	2.471	2.471
p3	NO	10^{-6}	4/2	2.479	3.629	2.466	2.466	2.472	2.472
p3	p1	10^{-3}	4/2	3.292	3.846	2.007	4.039	1.594	2.799
p3	p1	10^{-4}	4/2	3.301	3.847	2.006	4.027	2.821	4.022
p3	p1	10^{-5}	4/2	3.302	3.847	2.006	4.025	2.969	4.050

V_m	I_{ion}	Δt	Exp.	$V_m^{L_2}$	$V_m^{G_2}$	$u_1^{L_2}$	$u_1^{G_2}$	$u_2^{L_2}$	$u_2^{G_2}$
p3	p1	10^{-6}	4/2	3.302	3.847	2.006	4.025	2.983	4.051
p3	p3	10^{-3}	4/4	3.112	3.868	1.993	3.958	1.729	3.099
p3	p3	10^{-4}	4/4	3.108	3.869	1.993	3.955	2.546	3.961
p3	p3	10^{-5}	4/4	3.108	3.869	1.993	3.954	2.644	3.976
p3	p3	10^{-6}	4/4	3.108	3.869	1.993	3.954	2.653	3.976

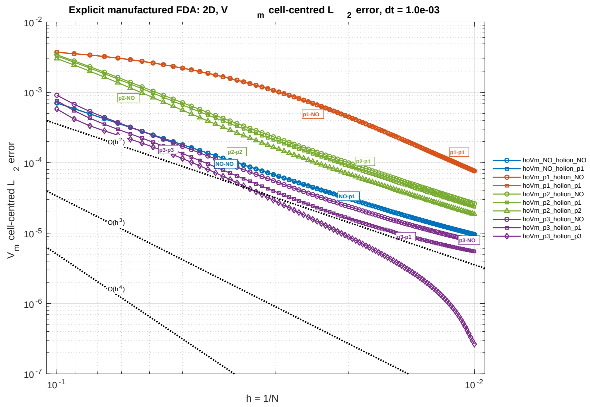
7.1 3D Figures

8 Convergence Figures

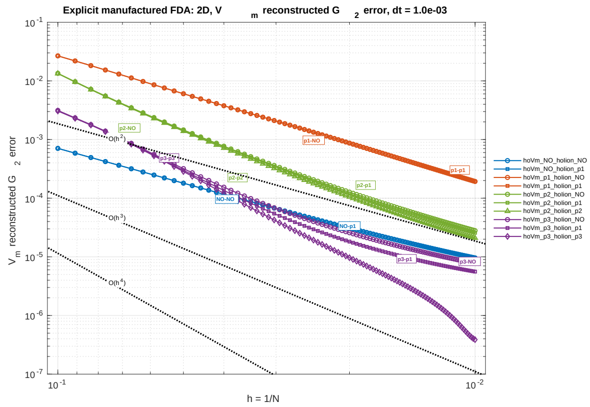
Run `plot_convergence.m` from this report directory to regenerate the PDF and PNG figures. The script scans both `results/example0/2D` and `results/example0/3D` for $\Delta t = 10^{-3}$, 10^{-4}, 10^{-5}, and 10^{-6}. It generates individual plot files, and the report assembles them into one six-panel figure for each dimension and time step. In each figure, rows correspond to V_m, u_1, and u_2, while the two columns compare cell-centred L_2 and reconstructed quadrature G_2 errors. The legend is placed outside each plot so that it does not cover the convergence curves. For a fixed dimension, variable, and error metric, the same axis limits and the same reference-line placement are used for all four time steps, so the Δt dependence can be compared directly. The horizontal axis is reversed so refinement proceeds from left to right, and the plotted $h = 1/N$ range is taken from the actual meshes available in each dimension. Dotted guide lines show $O(h^2)$, $O(h^3)$, and $O(h^4)$.

9 Notes

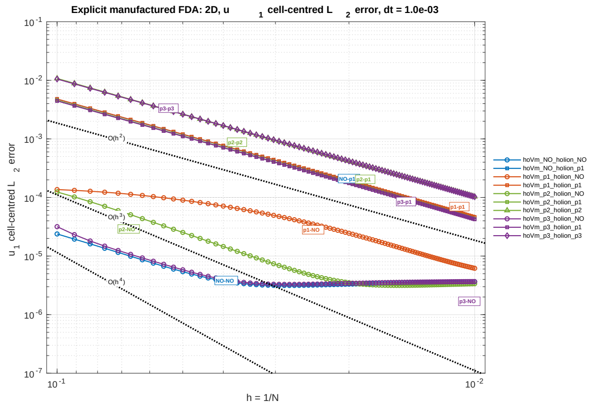
The all-mesh fitted order is sensitive to temporal error and pre-asymptotic meshes. This is visible in the state variables, especially for larger Δt. Comparing the rows across Δt separates spatial behaviour from forward-Euler time error. The reconstructed quantities V_m^G, u_1^G, and u_2^G should be interpreted as quadrature-point/reconstructed diagnostics, while V_m, u_1, and u_2 are the cell-centred values used by the convergence scripts.



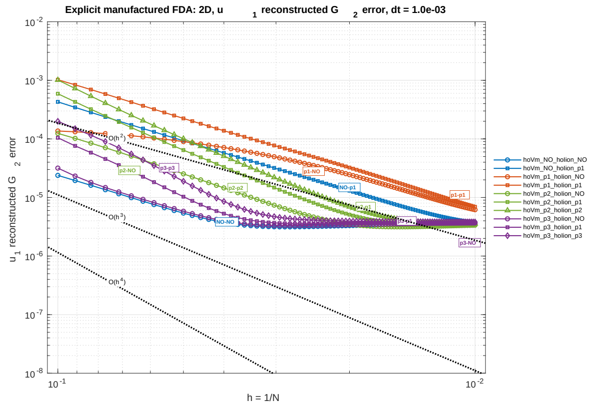
(a) $\|V_m\|_{L_2}$



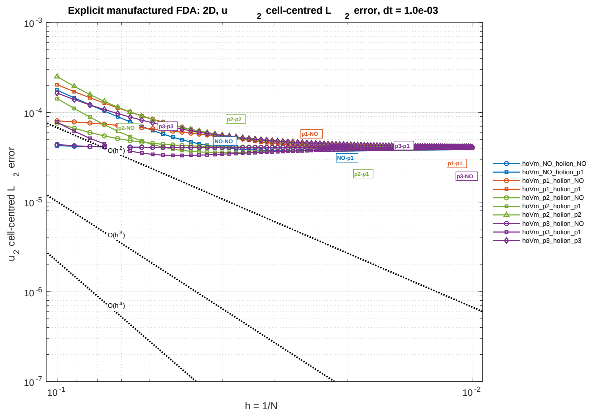
(b) $\|V_m\|_{G_2}$



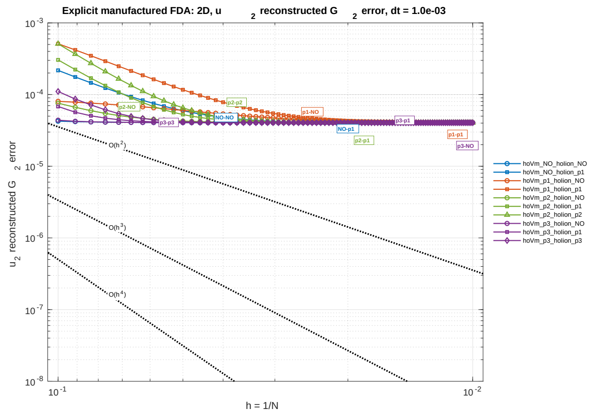
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

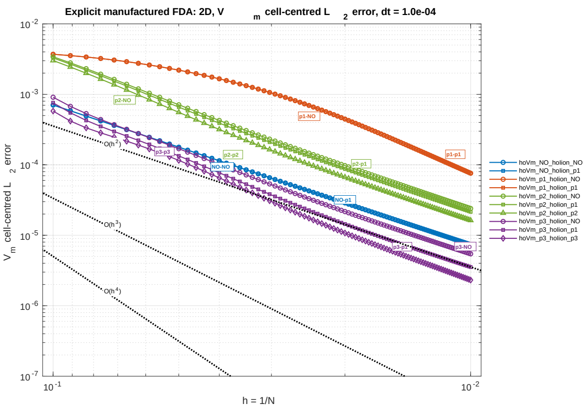


(e) $\|u_2\|_{L_2}$

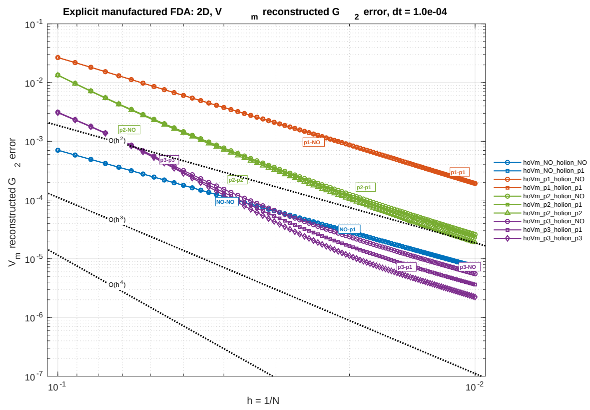


(f) $\|u_2\|_{G_2}$

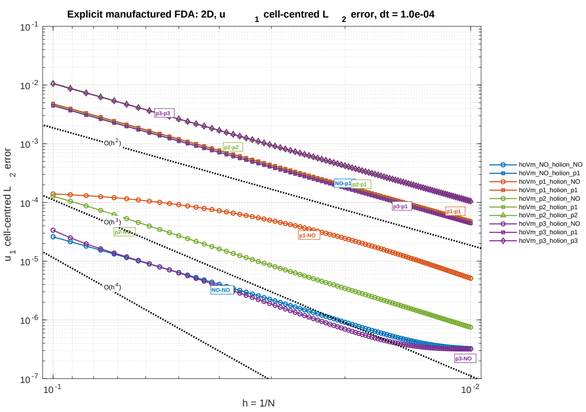
Figure 1: Explicit manufactured FDA solver, 2D results for $\Delta t = 10^{-3}$. Each row corresponds to one variable: V_m, u_1, and u_2 from top to bottom. The left column shows the cell-centred L_2 error and the right column shows the reconstructed quadrature G_2 error.



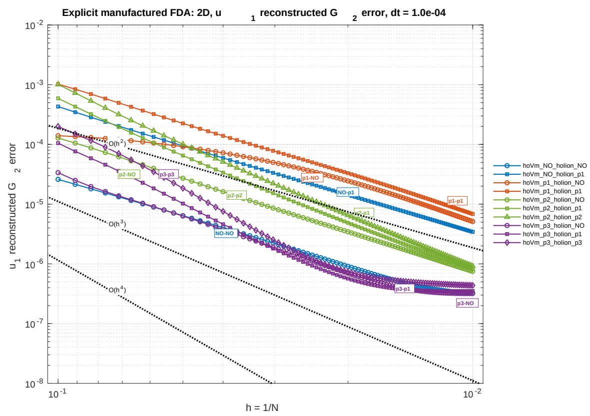
(a) $\|V_m\|_{L_2}$



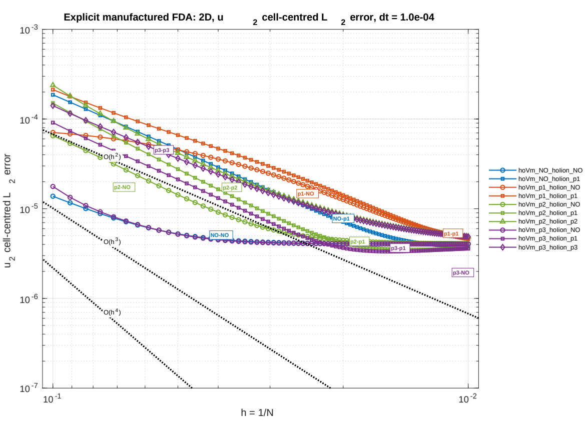
(b) $\|V_m\|_{G_2}$



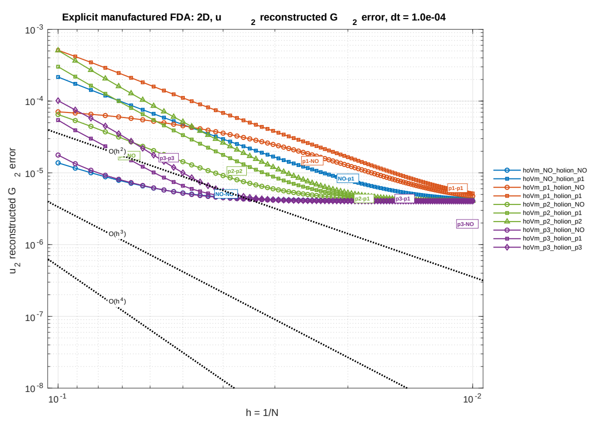
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

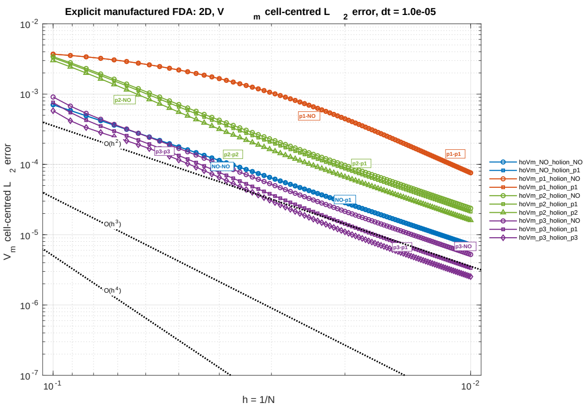


(e) $\|u_2\|_{L_2}$

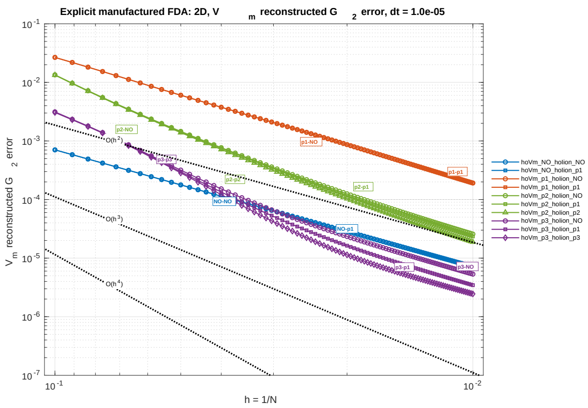


(f) $\|u_2\|_{G_2}$

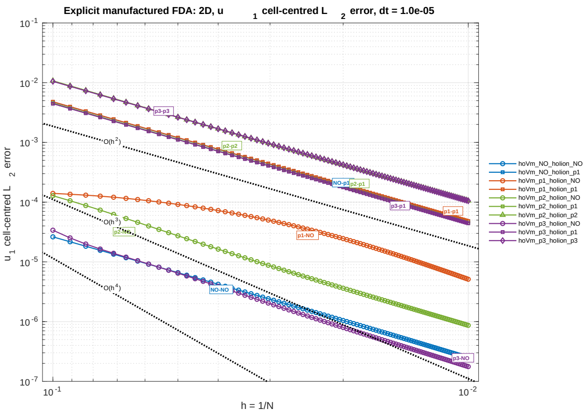
Figure 2: Explicit manufactured FDA solver, 2D results for $\Delta t = 10^{-4}$. Each row corresponds to one variable: V_m, u_1, and u_2 from top to bottom. The left column shows the cell-centred L_2 error and the right column shows the reconstructed quadrature G_2 error.



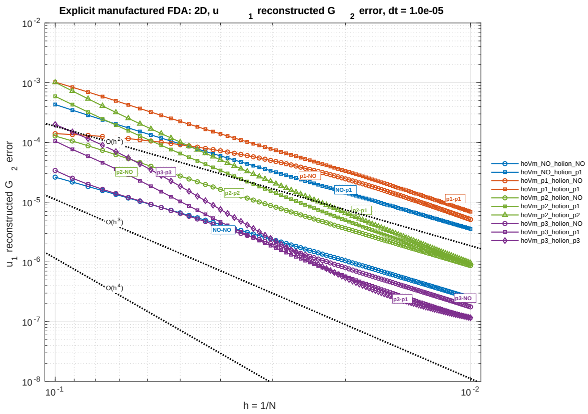
(a) $\|V_m\|_{L_2}$



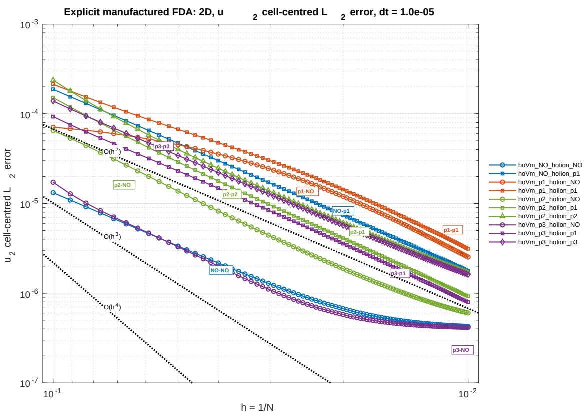
(b) $\|V_m\|_{G_2}$



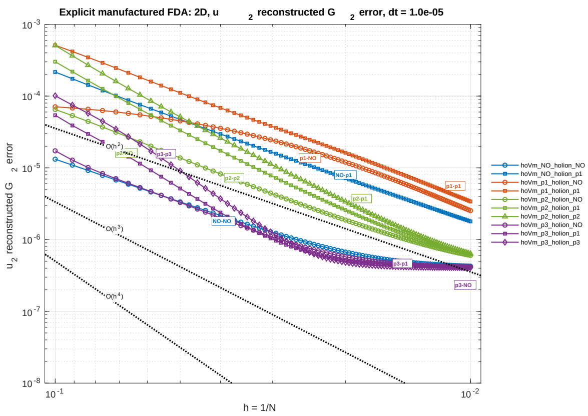
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

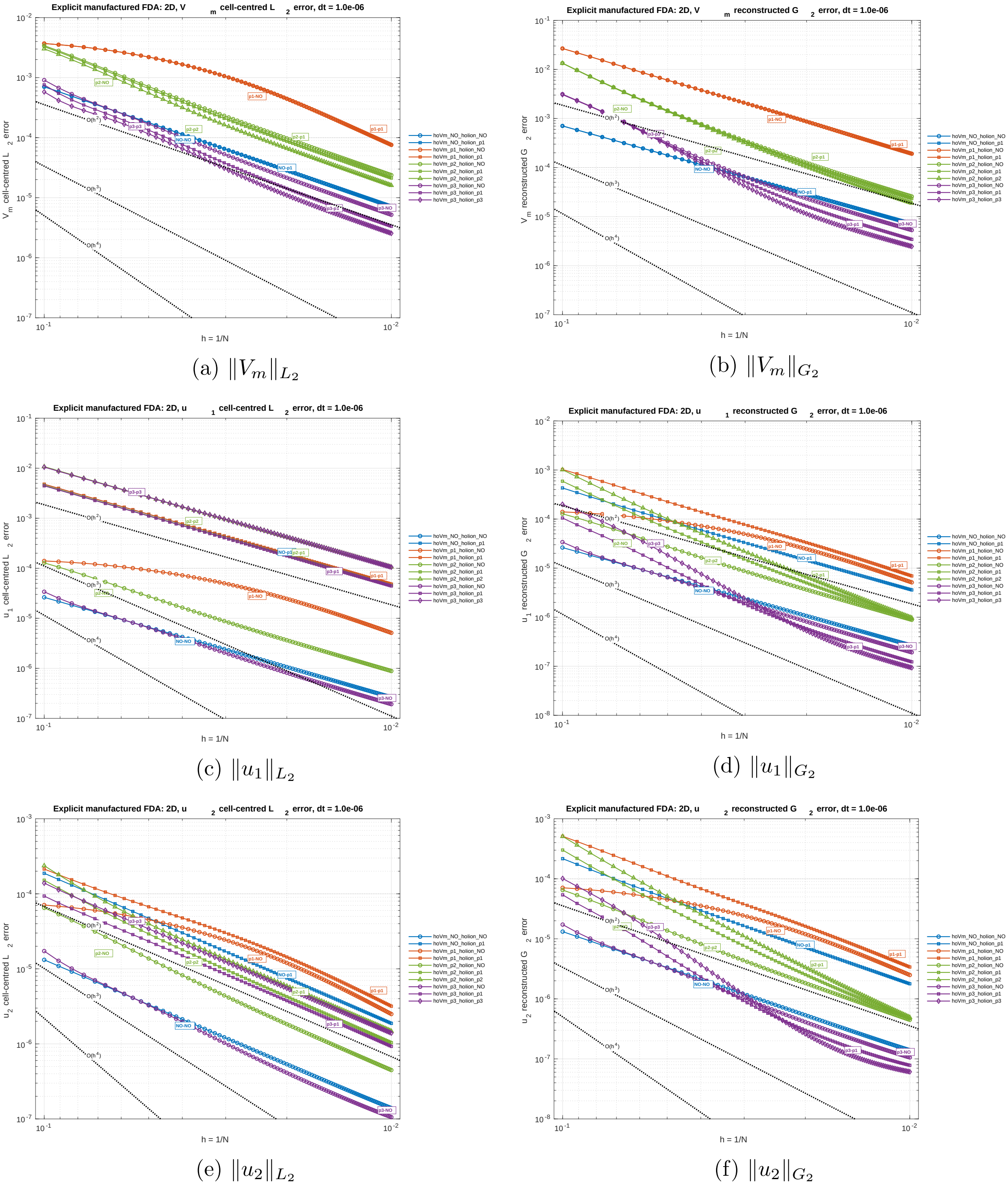


(e) $\|u_2\|_{L_2}$



(f) $\|u_2\|_{G_2}$

Figure 3: Explicit manufactured FDA solver, 2D results for $\Delta t = 10^{-5}$. Each row corresponds to one variable: V_m, u_1, and u_2 from top to bottom. The left column shows the cell-centred L_2 error and the right column shows the reconstructed quadrature G_2 error.



(a) $\|V_m\|_{L_2}$

(b) $\|V_m\|_{G_2}$

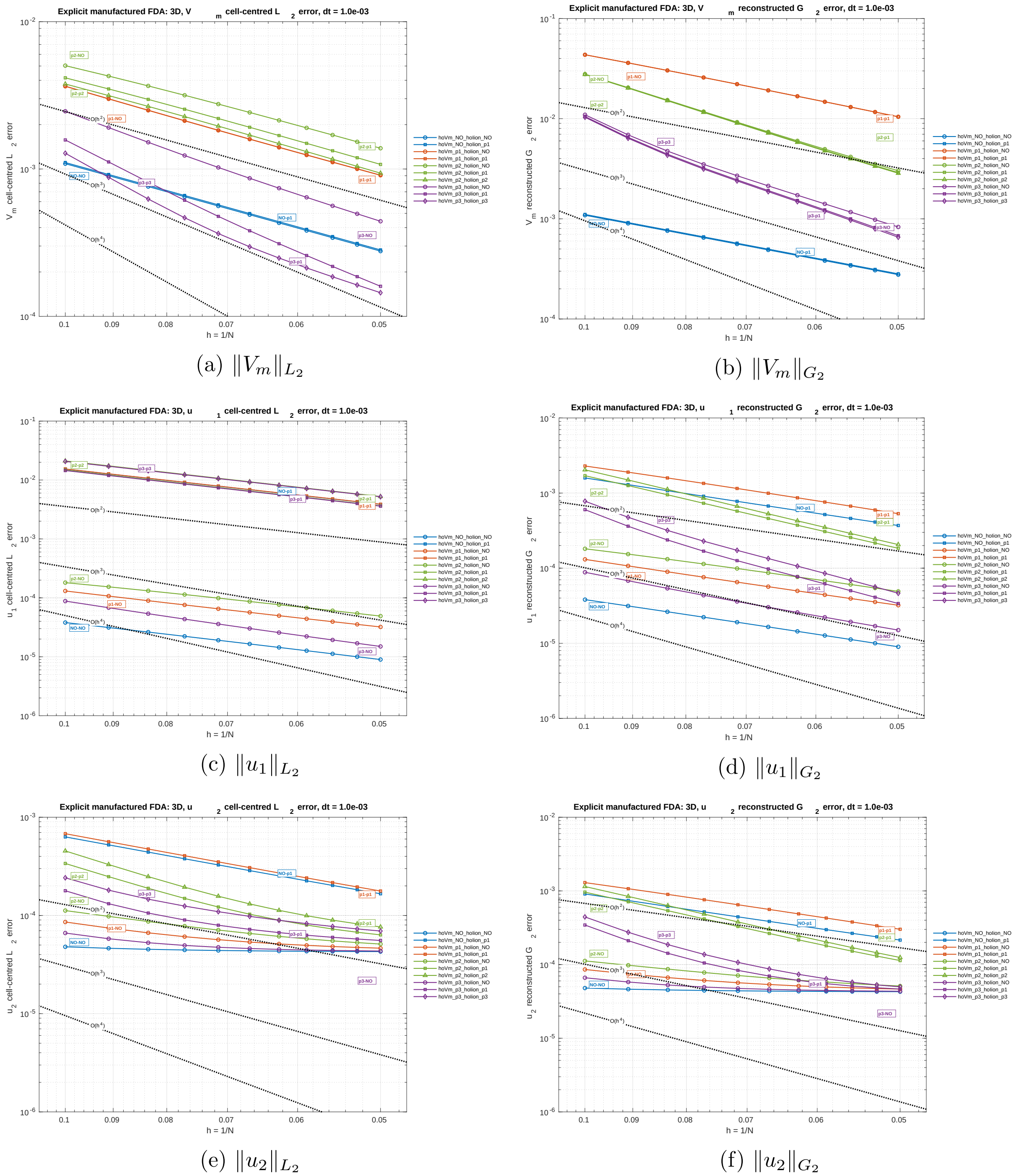
(c) $\|u_1\|_{L_2}$

(d) $\|u_1\|_{G_2}$

(e) $\|u_2\|_{L_2}$

(f) $\|u_2\|_{G_2}$

Figure 4: Explicit manufactured FDA solver, 2D results for $\Delta t = 10^{-6}$. Each row corresponds to one variable: V_m, u_1, and u_2 from top to bottom. The left column shows the cell-centred L_2 error and the right column shows the reconstructed quadrature G_2 error.

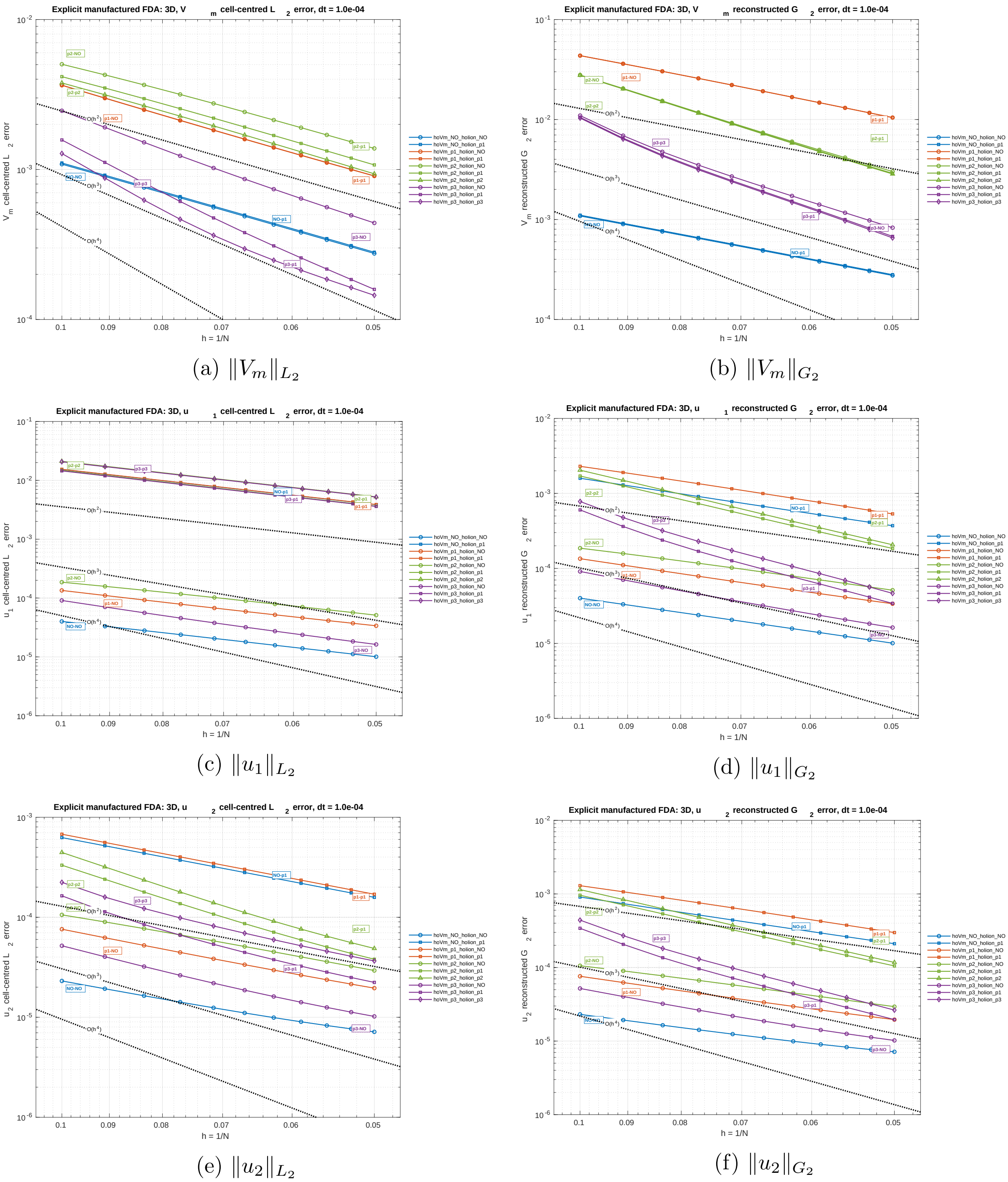


(a) $\|V_m\|_{L_2}$ (b) $\|V_m\|_{G_2}$

(c) $\|u_1\|_{L_2}$ (d) $\|u_1\|_{G_2}$

(e) $\|u_2\|_{L_2}$ (f) $\|u_2\|_{G_2}$

Figure 5: Explicit manufactured FDA solver, 3D results for $\Delta t = 10^{-3}$. Each row corresponds to one variable: V_m, u_1, and u_2 from top to bottom. The left column shows the cell-centred L_2 error and the right column shows the reconstructed quadrature G_2 error.



(a) $\|V_m\|_{L_2}$

(b) $\|V_m\|_{G_2}$

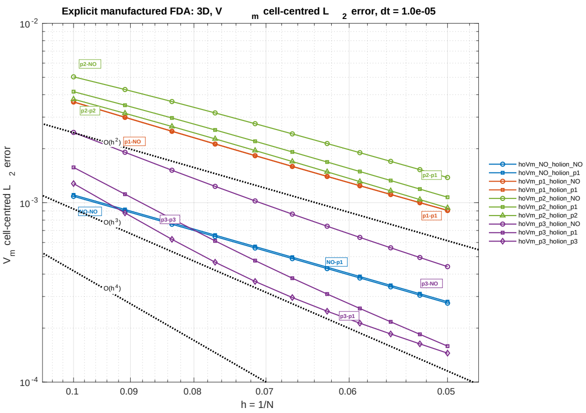
(c) $\|u_1\|_{L_2}$

(d) $\|u_1\|_{G_2}$

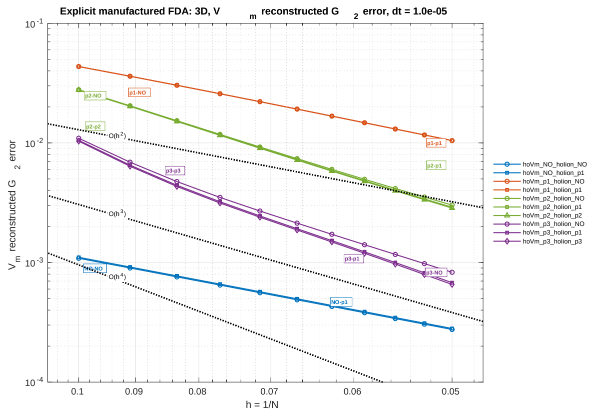
(e) $\|u_2\|_{L_2}$

(f) $\|u_2\|_{G_2}$

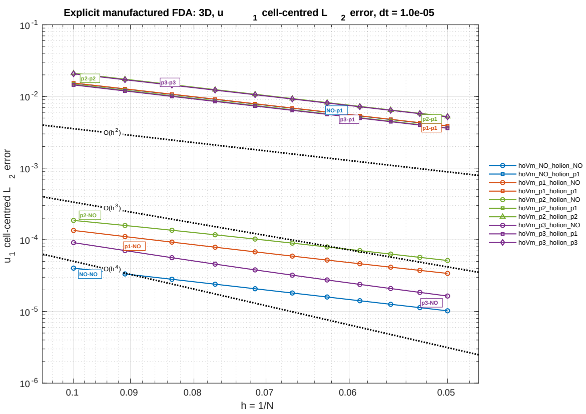
Figure 6: Explicit manufactured FDA solver, 3D results for $\Delta t = 10^{-4}$. Each row corresponds to one variable: V_m, u_1, and u_2 from top to bottom. The left column shows the cell-centred L_2 error and the right column shows the reconstructed quadrature G_2 error.



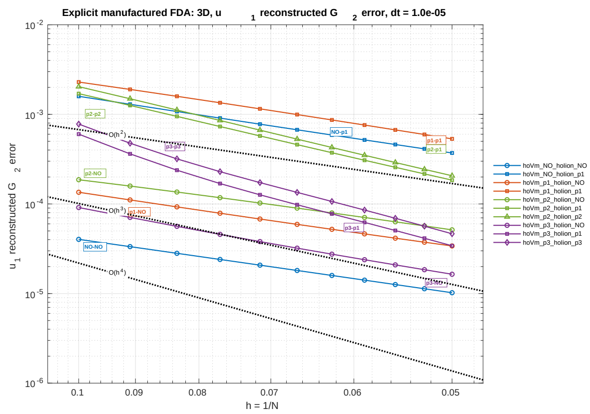
(a) $\|V_m\|_{L_2}$



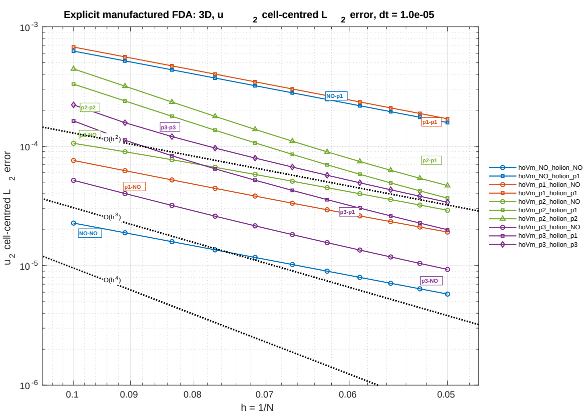
(b) $\|V_m\|_{G_2}$



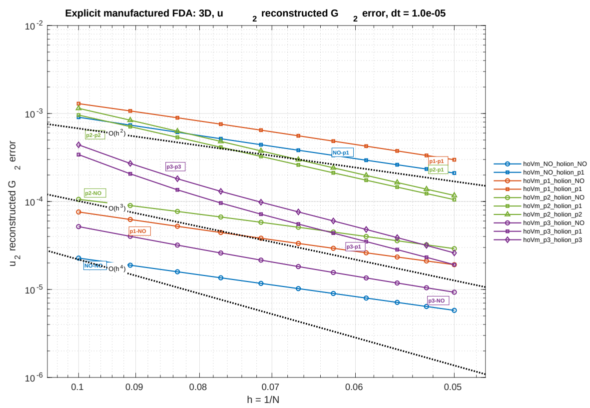
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$

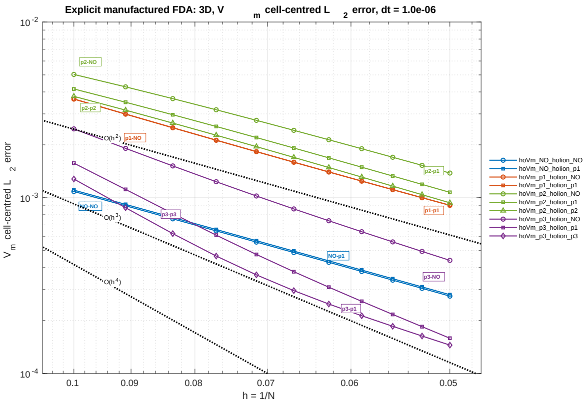


(e) $\|u_2\|_{L_2}$

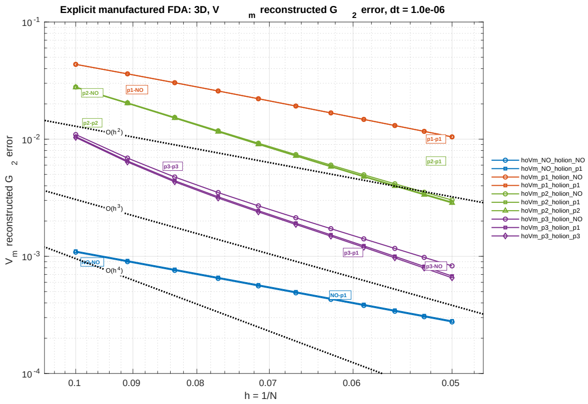


(f) $\|u_2\|_{G_2}$

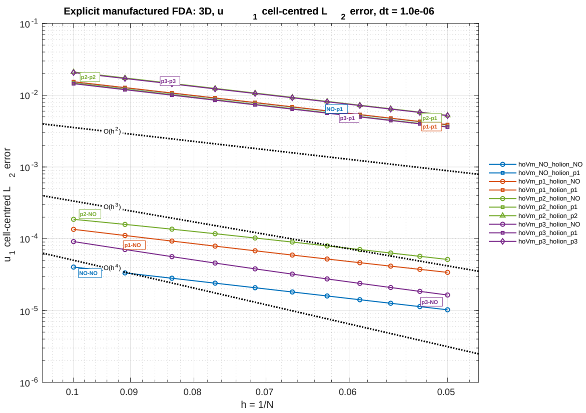
Figure 7: Explicit manufactured FDA solver, 3D results for $\Delta t = 10^{-5}$. Each row corresponds to one variable: V_m, u_1, and u_2 from top to bottom. The left column shows the cell-centred L_2 error and the right column shows the reconstructed quadrature G_2 error.



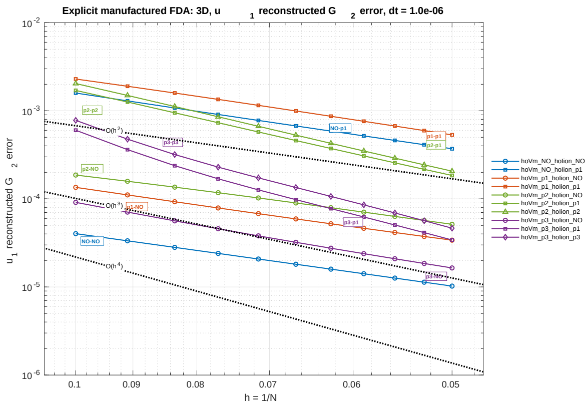
(a) $\|V_m\|_{L_2}$



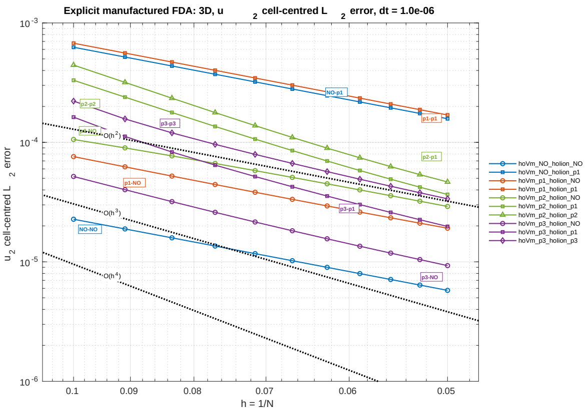
(b) $\|V_m\|_{G_2}$



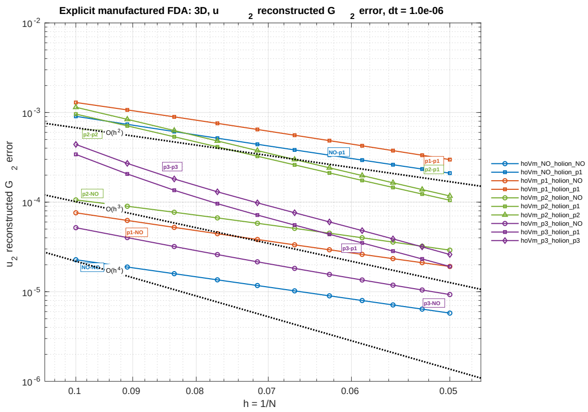
(c) $\|u_1\|_{L_2}$



(d) $\|u_1\|_{G_2}$



(e) $\|u_2\|_{L_2}$



(f) $\|u_2\|_{G_2}$

Figure 8: Explicit manufactured FDA solver, 3D results for $\Delta t = 10^{-6}$. Each row corresponds to one variable: V_m, u_1, and u_2 from top to bottom. The left column shows the cell-centred L_2 error and the right column shows the reconstructed quadrature G_2 error.